



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Unit 3 Lesson 1: Projectile motion

Class Outline

Title of Unit	Projectile and circular motions
Title of Lesson	Projectile motion
Starter Activity	Revision: quantities to describe linear motion
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	Demonstration: free-fall and projectile motion are the same motion at vertical direction Lecture: example of projectile motion Inquiry: Apply knowledge of projectile motion to predict the result of a simulation
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Put students in different groups and complete between groups to predict result in Simulation
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: Understand that the components of projectile motion can be considered separately. Understand that the horizontal component has constant velocity and the vertical component constant acceleration. Apply the suvat equations to the two components of the motion; Solve problems involving projectiles.
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication research
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Thinker, inquirer
Teaching Activities	Demonstration: freefall and projectile PHET simulation: firing a canon; group competition to guess the result of the simulation Video – ball launched from a moving cart
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Hint for homework question: estimate the pollution volume

Vocabulary



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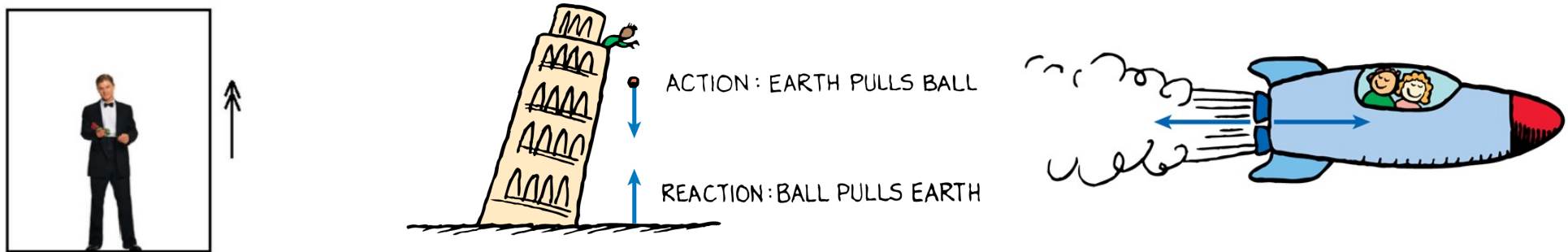
- Dimension 尺寸/维度
- Trajectory 轨迹
- Launch 发射
- Projectile 抛射
- Horizontal 水平的
- Vertical 竖直的
- Component 成分/组成部分
- Resolution of a vector 矢量分解

One dimensional motions



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So far we've been looking at motions along a straight line, or, one dimensional



Question: what physical quantity are used to describe a linear motion?

$F = ma$ (Newton's Second Law of Motion)

F in the is net force. a will take the same direction of net force

$v = u + at$ (kinematics)

Where v is the final velocity and u is the initial velocity.

What if net force (hence a) is not along the same line as v ?



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What shape do you think the trajectories of the balls will take?

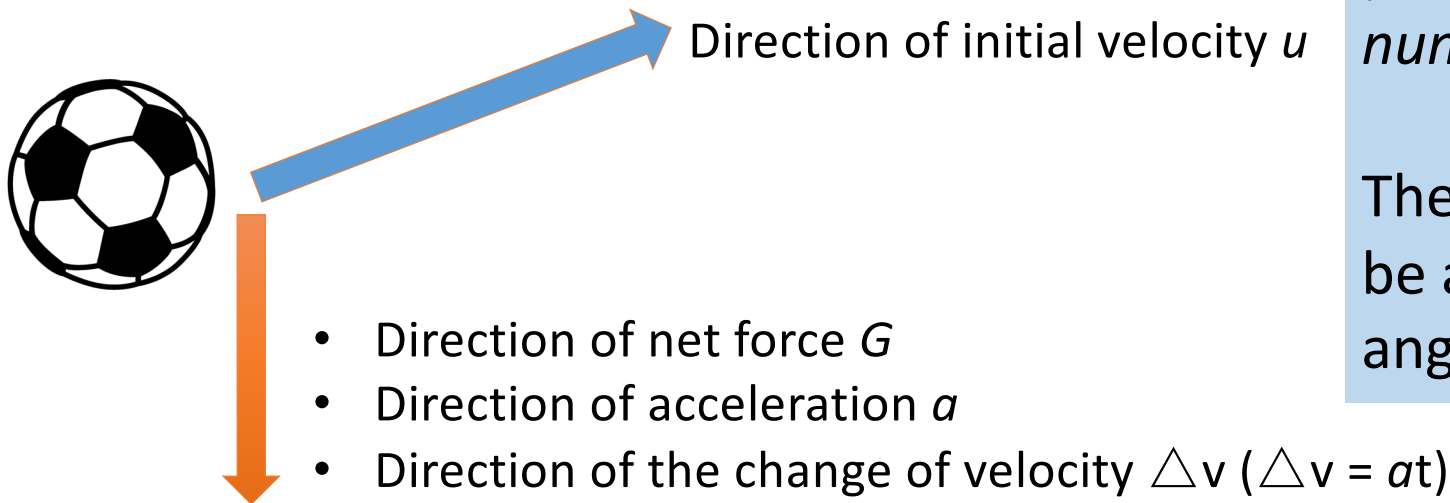
How do you use physical quantity to describe the motion of the moving ball in the picture?

What if the net force (hence a) is not along the same line as v ?



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Let's take a look at a flying ball in the air.



$v = u + at = u + \Delta v$
(add them as vectors, not numbers)

The resulting vector v will be a vector that is at an angle of u

FYI we will ignore air resistance in the discussion that follows...

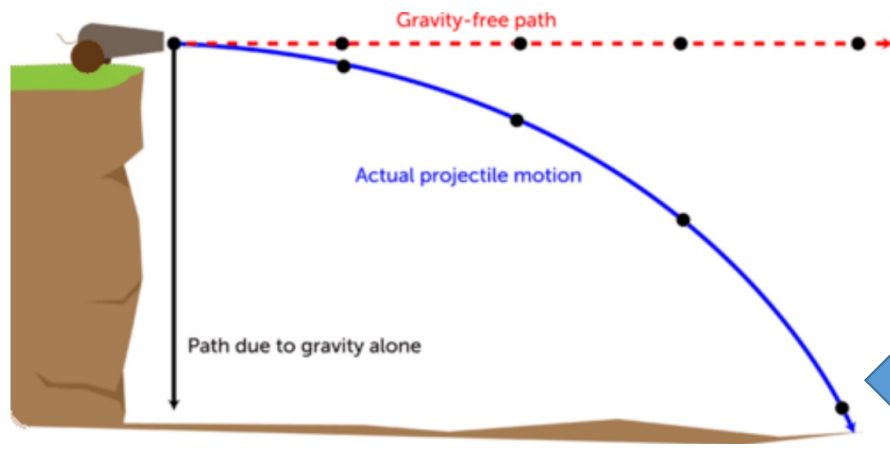
Projectile motion



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An object will move following a curved trajectory if the net force acting on it is not along the same line as its velocity.

Let's start with an easy example of such motion - Projectile Motion for an object launched horizontally



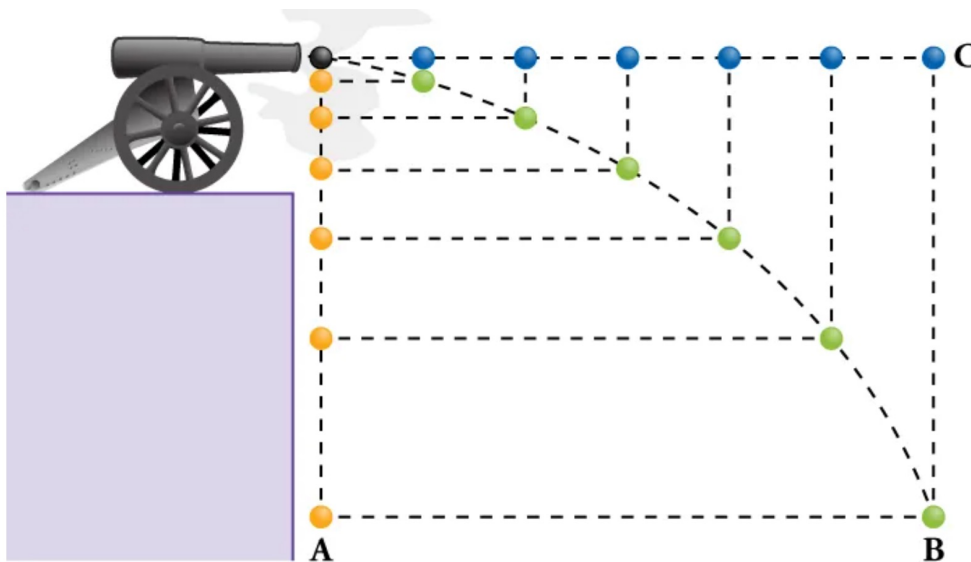
- Velocity at each moment will be along the tangent line of the trajectory
- Velocity at each moment can be resolved into x and y component
- Force along the x direction will only impact v_x
- Force along the y direction will only impact v_y
- Along each direction, $F = ma$ applies

View this as a horizontal uniform motion + vertical freefall motion

Projectile motion



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View this as a horizontal uniform motion
+
vertical freefall motion

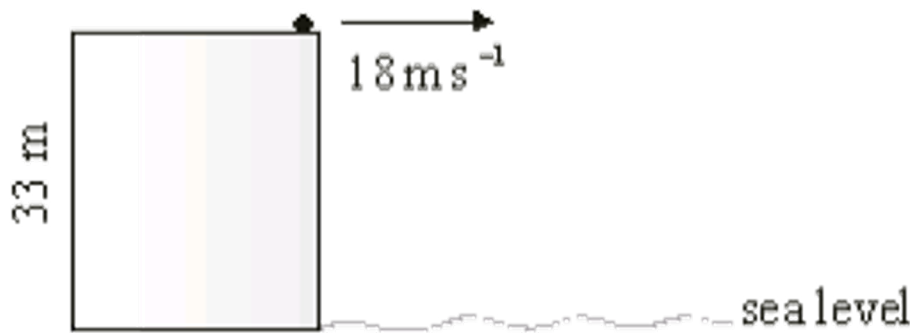
- Displacement at any time is the vector sum of horizontal displacement and vertical displacement
- Velocity at any time is the vector sum of horizontal velocity and vertical velocity

Projectile motion - example



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A stone is thrown horizontally from the top of a vertical cliff of height 33 m as shown below. The initial velocity of the stone was 18 m s^{-1} . Air resistance is negligible. ($g = 9.8 \text{ m s}^{-2}$)



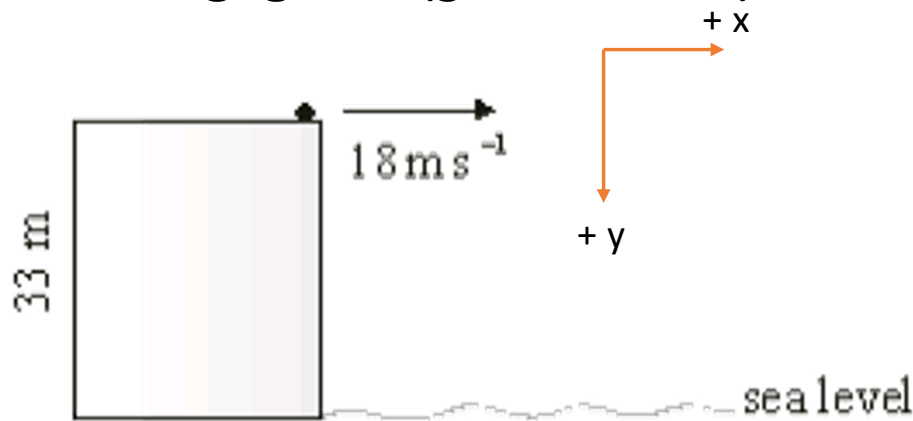
- State values for the horizontal and vertical acceleration of the stone
- Determine the time taken for the stone to reach sea level
- Calculate the distance of the stone from the bottom of the cliff when it reaches sea level
- Calculate the angle that the velocity makes with the surface of the sea

Projectile motion - example



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A stone is thrown horizontally from the top of a vertical cliff of height 33m as shown below. The initial velocity of the stone was 18 m s^{-1} . Air resistance is negligible. ($g = 9.8 \text{ m s}^{-2}$)



- a) State values for the horizontal and for vertical acceleration of the stone

$a_x = 0$ (horizontal net force is 0)

$a_y = g \approx 9.8 \text{ m s}^{-2}$ pointing downward (vertically, it's a free-fall motion)

- b) Determine the time taken for the stone to reach sea level

Vertically, $S = ut + \frac{1}{2}at^2$

$$33 = 0 \times t + 0.5 \times 9.8 \times t^2$$

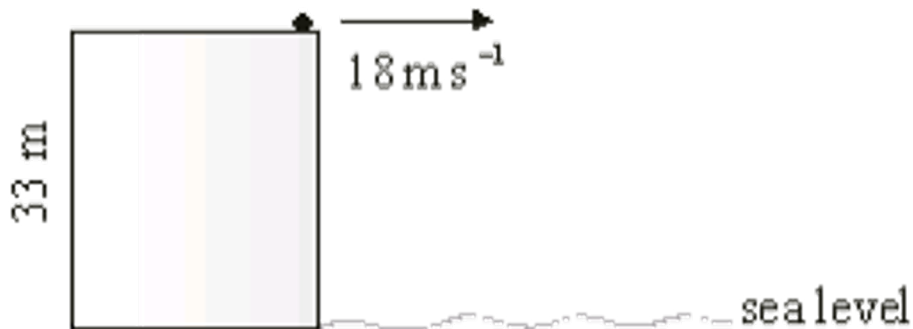
$$t = 2.6 \text{ s}$$

Projectile motion - example



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- c) Calculate the distance between the stone and the bottom of cliff when it reaches sea level

Horizontally: $S = vt$
 $S = 18 \times 2.6 = 47\text{m}$

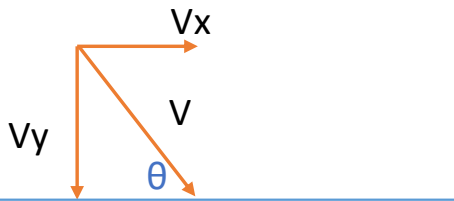
- d) Calculate the velocity when the stone reaches sea level, and the angle that the velocity makes with the surface of the sea

When reaching sea level:

$$v_x = 18\text{m/s}, v_y = gt = 9.8 \times 2.6 = 25.5 \text{ m s}^{-1};$$

$$v^2 = v_x^2 + v_y^2 \rightarrow v = 31.2 \text{ m s}^{-1};$$

$$\tan \theta = v_y / v_x = 25.5/18 \rightarrow \theta \approx 55^\circ$$

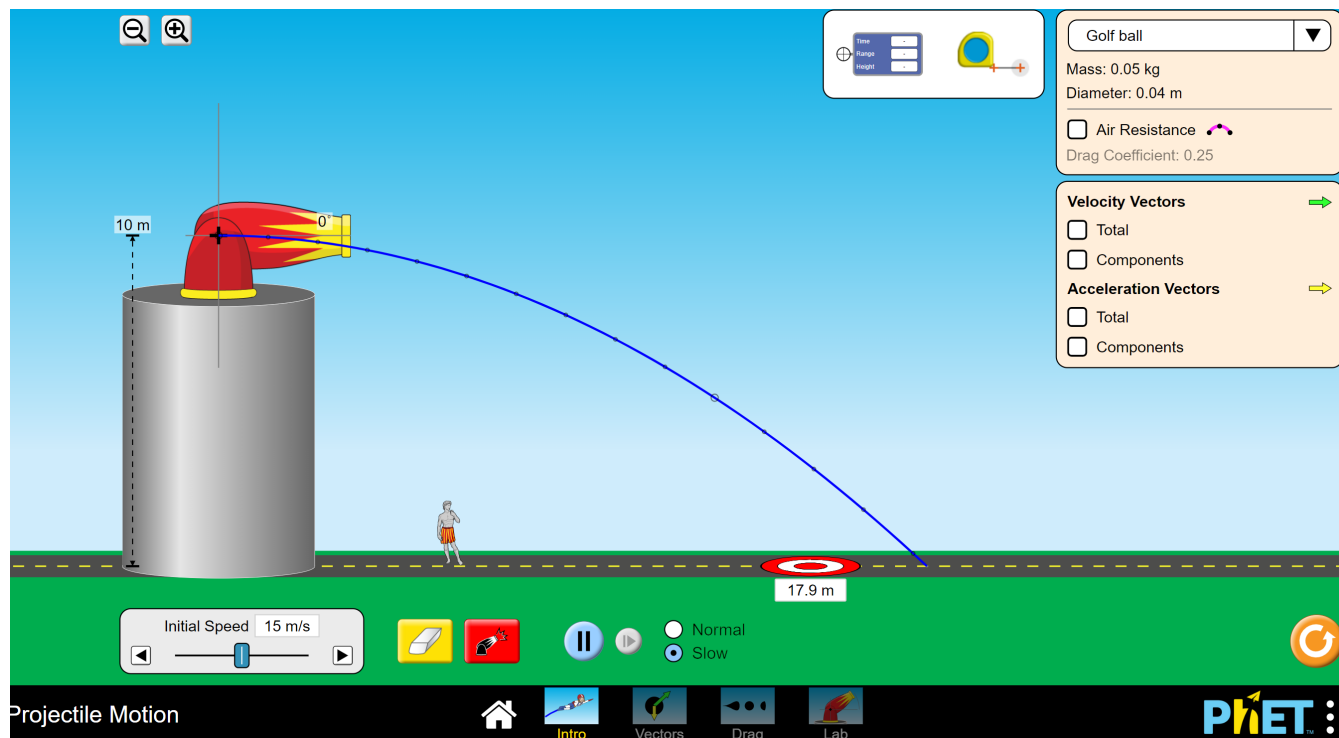


Projectile motion – mini competition



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Phet simulator: https://phet.colorado.edu/sims/html/projectile-motion/latest/projectile-motion_en.html



Projectile motion – mini competition



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1. If we choose to neglect air resistance, when I fire a golf ball and a car, will the trajectories be different?

**2. Height of platform = 12m
fire horizontally with an initial speed of 15m/s
neglect air resistance
How far will the cannonball travel horizontally?**

Projectile motion – mini competition



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3. Height of platform = 12m

fire horizontally with an initial speed of 15m/s

neglect air resistance

When the horizontal distance between the cannonball and the platform is 15m, how high is the cannonball above ground?

4. Height of platform is lowered to 10m

fire horizontally with an initial speed of 20m/s

When the ball is 5.1m above the ground,

What is the horizontal distance between the cannonball and the platform

Projectile motion – launch with an angle



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Let's move to something a little bit more complex - Projectile Motion for an object launched at an angle.

A cannonball is fired with an initial velocity of 100.0 m s^{-1} at an angle of 45° above the horizontal. What maximum height will it reach and how far will it fly horizontally?

Projectile motion – launch with an angle

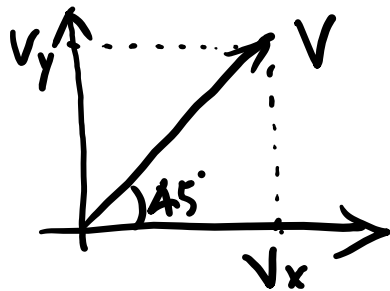


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Let's move to something a little bit more complex - Projectile Motion for an object launched at an angle.

A cannonball is fired with an initial velocity of 100.0 m/s at an angle of 45° above the horizontal. What maximum height will it reach and how far will it fly horizontally?

initial velocity $V = 100.0 \text{ m/s}$



$$V_x = V_y = V \cos 45^\circ \approx 70.7 \text{ m/s}$$

vertically, at max. height, $v = 0$, $v^2 - u^2 = 2as$

$$0 - 70.7^2 = 2 \times (-9.8) \times S, \quad \underline{S_y = 255 \text{ m}}$$

when landing, $S_y = 0 = ut + \frac{1}{2}at^2 \Rightarrow 70.7 \times t + \frac{1}{2} \times (-9.8) \times t^2 = 0, t = 14.4 \text{ s}$

$$S_x = V_x t = 70.7 \times 14.4 = \underline{1020 \text{ m}}$$